

Deploying Distilled AI Models for Email Phishing Detection

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Abstract & Introduction

Email Phishing remains one of the most significant cybersecurity threats, costing organizations billions annually. While large pre-trained language models like BERT demonstrate superior accuracy in detecting sophisticated phishing content, their substantial computational requirements make them impractical for real-time email filtering in production environments.

This paper presents a comprehensive framework for deploying knowledge-distilled AI models in enterprise email phishing detection systems. We propose an optimized knowledge distillation approach that transfers detection capabilities from large BERT models to compact TinyBERT architectures, enabling high-accuracy real-time phishing detection while maintaining computational efficiency.

Objectives

- To propose a phishing detection-specific knowledge distillation framework that enables lightweight transformer models to achieve accuracy comparable to large-scale BERT architectures while significantly reducing computational overhead
- To introduce a hybrid detection paradigm that integrates transformer-based natural language representations with handcrafted cybersecurity features for improved robustness against diverse and evolving phishing attacks.

Method

Our stacked classifier fuses TinyBERT embeddings with URL, header, and keyword features via attention-based fusion, followed by a LightGBM ensemble with confidence calibration. The distillation process uses attention transfer, response-based distillation ($T=2.5$), progressive compression through intermediate models, and task-specific adaptation for URL and sender scoring. The framework is deployed as a containerized microservice using FastAPI, Docker, and ONNX Runtime with Kubernetes auto-scaling.

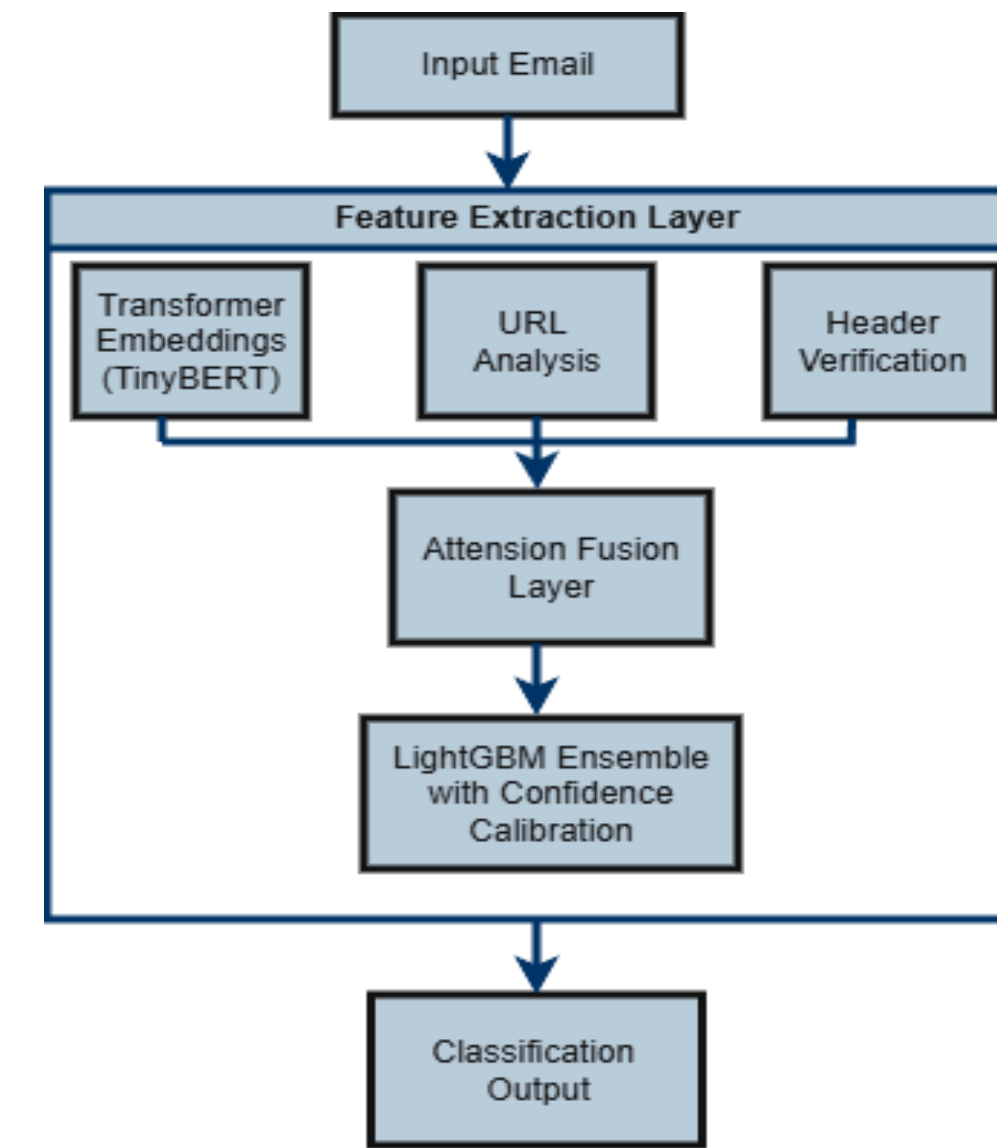
Methodology

Our system employs a stacked classifier architecture that fuses distilled transformer embeddings with traditional cybersecurity features through attention-based mechanisms. The TinyBERT student model (distilled from BERT-base via progressive distillation) extracts semantic features, while parallel processing extracts URL, header, and keyword features. These are combined via attention fusion and passed to a LightGBM ensemble with confidence calibration for final classification.

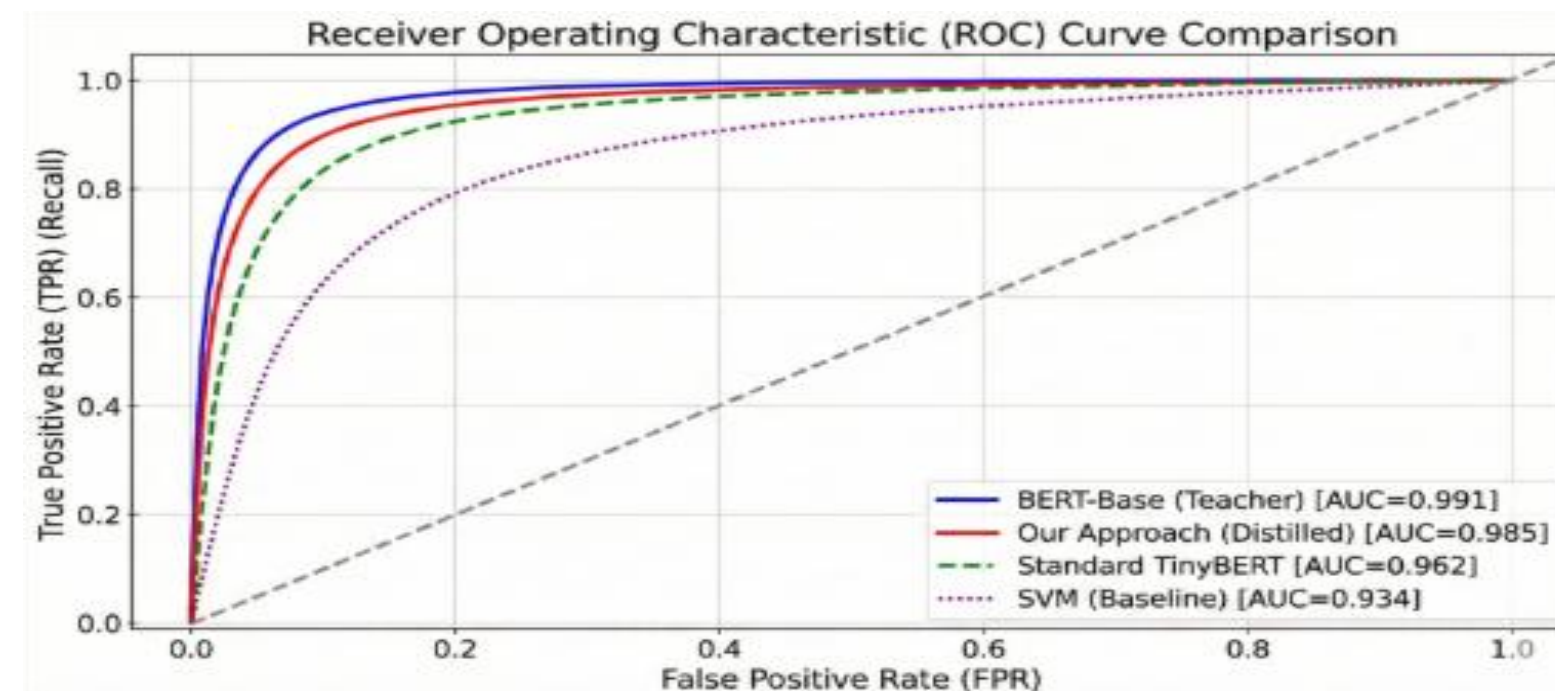
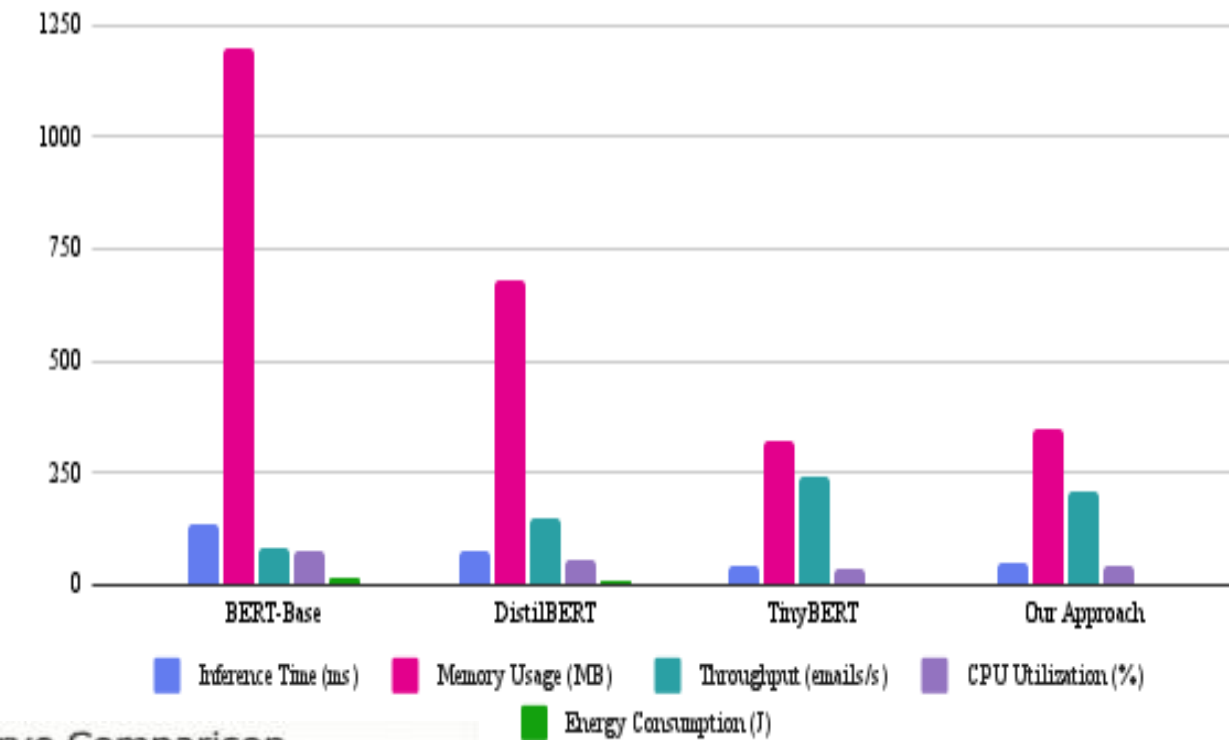
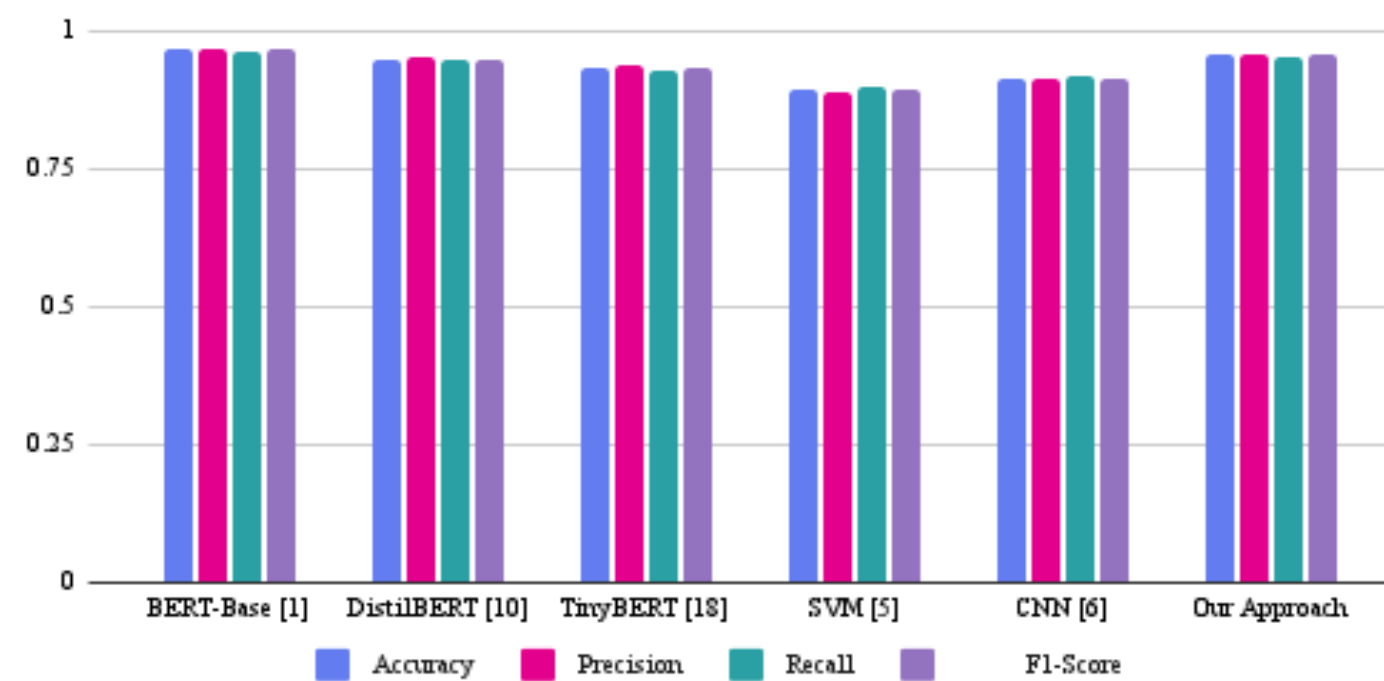
Data Processing: Email parsing, URL analysis (VirusTotal), header verification (SPF/DKIM/DMARC), feature extraction

Knowledge Distillation: Progressive compression BERT-base $\rightarrow$ TinyBERT with attention transfer ($T=2.5$)

Hybrid Classification: Attention-based fusion of transformer outputs + security features



Result



Result (If extra space needed)

Our distilled TinyBERT framework achieves enterprise-ready phishing detection with 95.7% accuracy while reducing:

- Model size by **85%** (440MB $\rightarrow$ 65MB)
- Inference latency by **61%** (135ms $\rightarrow$ 52ms)
- Energy consumption by **84%** (12.5J $\rightarrow$ 4.8J)

Throughput increases by **147%** (85 $\rightarrow$ 210 emails/sec) with balanced errors (FPR: 4.1%, FNR: 4.6%).

Conclusion & Future Work

Our approach achieves 95.7% detection accuracy while reducing model size by 85% and inference latency by 61%, establishing a practical benchmark for resource-constrained security applications.

Future Work:

- Longitudinal studies in production environments across multiple organizations
- Enhanced detection of adversarial phishing techniques using generative AI
- Integration of user behavior analytics and email interaction patterns.

References

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Model	Accuracy(%)	F1-Score(%)
BERT-Base	96.5	96.5
DistilBERT	94.8	94.8
TinyBERT	93.2	93.1
Ours	95.7	95.6

Model	Time(ms)	Memory(MB)	Throughput(emails/s)
BERT-Base	135	1200	85
DistilBERT	75	680	150
TinyBERT	45	320	240
Ours	52	350	210